

# Model Effectiveness Evaluation Workbench

ELEC5623 Group Business Proposal candidate

Selected direction: **Track B - Product that helps AI work better**

Candidate status: **unapproved, incomplete, and not submission-ready**

Official assessment: 10% (10 marks), group assignment

Due: 8 September 2026, 23:59 through Canvas

Recommended length: 8-10 pages excluding only the cover and references

This compressed candidate follows the official 12-section structure. It does not claim group formation, stakeholder validation, tutor approval, a frozen evaluation result, or Canvas submission. Bracketed fields and **GroupXX** are deliberate blocking sentinels. They must be replaced with verified facts before rebuilding or renaming this candidate as the submission PDF. The group must count every page other than the cover and reference-only pages; a page shared by Section 11 and references remains a body page. No appendix is assumed to be outside the page recommendation. The 8-10 pages are an official recommendation, but the rendered candidate treats that range as its internal page-QA pass condition.

## 1. Cover and group information

| Cover field                       | Candidate value                                                                                                                                                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project title                     | Model Effectiveness Evaluation Workbench                                                                                                                                                                                    |
| Selected direction                | Track B - Product that helps AI work better                                                                                                                                                                                 |
| Group number                      | GroupXX - [REQUIRED: actual Canvas group number]                                                                                                                                                                            |
| Full member names and student IDs | [REQUIRED: actual confirmed members and SIDs]                                                                                                                                                                               |
| Lab tutor                         | [REQUIRED: authorised lab tutor name]                                                                                                                                                                                       |
| Approval date                     | [REQUIRED: actual approval date]                                                                                                                                                                                            |
| Brief approval statement          | [REQUIRED: exact approved wording or verified concise statement]                                                                                                                                                            |
| Product vision                    | On the same evidence-backed tasks, run $\geq 2$ named models; report quality, task-fit, latency, and estimated cost separately; contrast min-cost vs quality/task-fit; human review remains success; not auto model-deploy. |

Tutor approval is mandatory before the deadline and must confirm relevance, originality, semester feasibility, and appropriate scope. A passing local test suite is not approval evidence. Suggested final filename, after the group gate is closed: **ELEC5623\_GroupXX\_Proposal.pdf**.

## 2. Executive summary

A student can ask two models to write up the same lab notebook. The useful question is which model hallucinates less and fits the stated task better, not whether cheaper means worse. Reviewers still need claim-level evidence, quotations, requirement coverage, and an auditable decision. Candidate users are students, evidence reviewers, requirement owners, authors, and team leads; these roles are hypotheses pending approved validation.

The proposed **Model Effectiveness Evaluation Workbench** runs  $\geq 2$  named models on one English JSON bundle of requirements, evidence, and AI-generated Markdown. The inner **Inspector** engine (FR-01-15) segments claims, retrieves local evidence, classifies support, enforces citations and exact quotes, maps requirements, and exports an immutable run. The compare shell (FR-16-18) reports quality, task fit, latency, and estimated list-price cost separately and contrasts **min-cost** with a **quality/task-fit** rule. No combined index; no auto model-deploy. A completed run means only **READY\_FOR\_HUMAN\_REVIEW**. This is independent ELEC5623 Track B coursework.

First-month scope is Python 3.11, FastAPI, Pydantic v2, scikit-learn, and pytest, with synthetic English JSON/Markdown only. Live commercial compare APIs are out of scope this week. Team allocation, stakeholder evidence, tutor approval of this Workbench direction, the annotation protocol, final thresholds, and frozen-corpus results remain open gates.

## 3. Introduction and background

The project addresses AI engineering evaluation: whether generated output uses a bounded evidence set and satisfies stated requirements. Whole-document scores can hide mixed support. FActScore motivates decomposing long-form generation into atomic facts because supported and unsupported content may coexist [4]. This proposal adopts that claim-level motivation only; it does not reuse the FActScore implementation, data, experiments, or metric. RAGAs separates aspects such as retrieval relevance and faithful use of retrieved information [5]. The Inspector similarly keeps retrieval matches, support labels, evidence-integrity checks, requirement mappings, and human decisions visible instead of collapsing them into one unexplained score. Token spend and model list price are not treated as quality or task-fit proxies. It does not implement the RAGAs suite.

The assumed current workflow is manual: identify claims, search supplied evidence, check citations and quotations, compare requirements, and record a decision. This workflow and its cost are hypotheses; no interview, time saving, error reduction, organisation, or market figure is claimed. A permitted stakeholder method or tutor-approved alternative must establish which review step is costly, which errors matter most, and which trace fields users need.

Alternative solution classes include manual review, document search, generic LLM critique prompts, research evaluation frameworks, and broad governance platforms. Manual review preserves authority but may vary in structure. Document search retrieves passages without assessing claims. Generic prompts are convenient but do not guarantee stable schemas, immutable inputs, or fail-closed behaviour. Existing research tools provide credible context but target different data, metrics, or workflows. The proposed contribution is a narrow, independent course product joining claim evidence, requirements, metrics, and human review in one reproducible contract.

## 4. Problem statement and motivation

The target decision is: **given the same stated requirements, supplied evidence, and outputs from at least two named models, which model hallucinates less / fits the task better, and should min-cost or quality/task-fit drive that choice?** The gap is not another generator. It is a bounded packet that links input hash, stable claim, evidence, support decision, mapped requirements, per-model quality/task-fit/time/estimated-cost, two named selections, and later human review.

Students checking a lab write-up risk missing invented numbers; reviewers risk overlooking weak claims; leads need reproducibility. These remain hypotheses until validated. GenAI is appropriate only for interpreting varied claim-evidence language. Deterministic code owns schema, ranking, citation and quote integrity, artifacts, and approval boundaries. Estimated cost is list-price  $\times$  conservative token bound, not an invoice. The product does not auto-deploy a model and does not claim that the quality rule predicts deployment success. No unverified time, financial, accuracy, or market claim is made.

## 5. Stakeholders and requirements engineering

### 5.1 Stakeholder needs and validation

| Hypothesised stakeholder | Need                                                        | Main design risk                             |
|--------------------------|-------------------------------------------------------------|----------------------------------------------|
| Evidence reviewer        | Locate support and challenge weak claims in one queue       | Automation bias or missed context            |
| Requirement owner        | See supported claim-level coverage and explicit gaps        | Lexical mapping overstates semantic coverage |
| Report author            | Correct unsupported, contradicted, or misquoted statements  | Writing to the tool rather than to evidence  |
| Team lead or assessor    | Reproduce inputs, configuration, output, and review history | Synthetic metrics create false confidence    |
| System administrator     | Operate a bounded, fail-closed evaluation path              | Data leakage or provider outage              |

The proposed primary stakeholders are students checking a lab write-up, evidence reviewers, requirement owners, and report authors. Proposed secondary stakeholders are team leads or assessors and system administrators. Roles and needs are hypotheses. No interview, survey finding, or tutor approval is claimed.

**Proposed tutor-approved alternative validation method** (status: proposed, not executed, awaiting written tutor agreement; [REQUIRED: written tutor agreement before any walkthrough]): a structured 20-minute walkthrough with the lab tutor as proxy reviewer, using `examples/daily_lab_writeup.json` and the `fixture vs fixture-b compare.md`. Record which review step is costly, whether missed-risky claims or false escalations are costlier, and which trace fields a decision record must retain. Identifiable notes stay private. Until that agreement is written and the walkthrough is run, every stakeholder need remains a hypothesis. The complete Lab 2 written use case is in Section 7.

### 5.2 Scope and priority

Must scope includes strict schemas and IDs, stable claim segmentation, local retrieval, five support labels, citation and exact-quote enforcement, requirement mapping and metrics, no-clobber artifacts, append-only review, CLI/API parity, sequential named-model compare with min-cost vs quality/task-fit, and fail-closed tests. Should scope includes frozen-set error analysis and clean-checkout rehearsal. Could scope includes lightweight report visualisation or a second synthetic domain. The first month will not include PDF/image/OCR input, web search, a database, a frontend framework, live commercial compare APIs, autonomous approval, or cross-project assessed artifacts.

### 5.3 Functional requirements

All 18 FRs are draft until the group and tutor confirm scope and allocation. The official minimum is three FRs per person. Eighteen FRs still allow at least three per person for a confirmed team of no more than five.

| ID    | Functional requirement                                               | Testable acceptance criterion                                            |
|-------|----------------------------------------------------------------------|--------------------------------------------------------------------------|
| FR-01 | Validate a versioned English JSON bundle with strict schemas.        | Invalid or extra fields fail before a run.                               |
| FR-02 | Reject duplicate requirement, evidence, and expected-claim IDs.      | Each duplicate category fails with no report.                            |
| FR-03 | Segment English Markdown prose/lists into stable claim IDs.          | Same input yields the same ordered IDs; heading/code-only fails.         |
| FR-04 | Retrieve ranked local evidence for every claim.                      | Each claim records bounded matches, scores, and excerpts.                |
| FR-05 | Use TF-IDF and identify any lexical fallback.                        | Each run records <code>sklearn-tfidf</code> or the named fallback.       |
| FR-06 | Produce exactly one of five support labels per claim.                | Output is one of the five-label enum values.                             |
| FR-07 | Extract evidence citations for each claim.                           | Citation IDs are explicit and traceable to the bundle.                   |
| FR-08 | Block positive support based on an unknown citation.                 | Unknown citations cannot remain positive.                                |
| FR-09 | Check quoted spans exactly against cited evidence.                   | An altered quote cannot retain positive support.                         |
| FR-10 | Map claims only to stated requirement IDs.                           | Unknown IDs are rejected; mappings stay in-bundle.                       |
| FR-11 | Calculate citation, quote, coverage, and label metrics.              | JSON and Markdown exports agree.                                         |
| FR-12 | Score only complete supplied truth.                                  | Non-empty <code>expected_claims</code> must cover segmented IDs exactly. |
| FR-13 | Export/read no-clobber artifacts.                                    | Aliased, partial, or mismatched reports fail.                            |
| FR-14 | Append reviews without rewriting evidence.                           | History/hash persist; alias targets fail unchanged.                      |
| FR-15 | Expose one Inspector service through CLI and HTTP.                   | CLI and API contract tests pass.                                         |
| FR-16 | Sequentially run $\geq 2$ named gateways on the same bundle.         | Each model yields an independent Inspector run.                          |
| FR-17 | Report quality, task-fit, time, estimated cost, and two selections.  | Min-cost vs quality/task-fit; no combined index.                         |
| FR-18 | If complete labels exist, check each policy vs annotation-preferred. | Same ranking keys as quality/task-fit; else comparison only.             |

The public CLI is `validate bundle.json`, `evaluate bundle.json --out runs/`, `compare bundle.json --out DIR --models fixture,fixture-b`, and `review RUN_ID review.json --runs runs/`. The HTTP surface is `GET /health`, `POST /v1/evaluations`, `GET /v1/evaluations/{run_id}`, and `POST /v1/evaluations/{run_id}/reviews`. Adapters call the same Inspector service rather than duplicating decision rules.

## 5.4 Non-functional requirements, constraints, and assumptions

| ID     | Quality and proposed measurable criterion                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Traceability: 100% of serialized assessments contain required claim, evidence, decision, requirement, configuration, and input-hash fields.              |
| NFR-02 | Reproducibility: two fixture runs produce byte-equivalent normalized reports.                                                                            |
| NFR-03 | Performance: 200 frozen claims complete in $\leq 60.0$ seconds on the nominated lab machine.                                                             |
| NFR-04 | Token budget: a conservative canonical UTF-8 byte bound is checked before provider use.                                                                  |
| NFR-05 | Fail closed: unsafe input, incomplete/mismatched truth, or provider failure creates no completed <code>report.json</code> .                              |
| NFR-06 | Privacy: only the minimum configured payload crosses an approved provider boundary.                                                                      |
| NFR-07 | Auditability: runs are no-clobber; report/review targets are unaliased; reviews are append-only.                                                         |
| NFR-08 | Onboarding: a new member reproduces the README flow within 20 minutes.                                                                                   |
| NFR-09 | Maintainability: core package branch coverage remains $\geq 80\%$ .                                                                                      |
| NFR-10 | Comparison latency: two-model fixture compare of the sample/daily bundle finishes in $\leq 60.0$ s; a failed model publishes no complete compare report. |

The 60-second, budget, onboarding, and coverage values are proposed engineering criteria, not marking thresholds. They remain subject to tutor feedback and measurement on the nominated environment.

| ID   | Design constraint and consequence                                                               |
|------|-------------------------------------------------------------------------------------------------|
| C-01 | Python $\geq 3.11$ , $< 3.12$ and fixed first-month libraries provide one bounded toolchain.    |
| C-02 | English JSON/Markdown only; no PDF, image, OCR, or multilingual parsing.                        |
| C-03 | Supplied local evidence only; no web retrieval or database.                                     |
| C-04 | <code>FixtureModelGateway</code> is CI truth; live-model output cannot replace reproducibility. |
| C-05 | Synthetic data only until provenance, privacy, licence, and tutor approval are recorded.        |
| C-06 | Every successful machine run stops at human review; no approval or external action.             |

Assumptions requiring validation are that users can supply a bounded bundle, English text is sufficient for the first milestone, the course environment can run Python 3.11, and a synthetic corpus is acceptable. A failed assumption triggers recorded scope review rather than silent expansion.

## 5.5 Requirement-to-evaluation trace

| Stakeholder need                                            | Priority | Linked requirements                       | Candidate acceptance evidence                                                                                            |
|-------------------------------------------------------------|----------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Review a claim with bounded evidence and no hidden approval | Must     | FR-04, FR-06-09, FR-13-14; NFR-01, NFR-07 | Claim trace contains ranked evidence, enforced citation/quote status, immutable report, and appended human decision.     |
| Identify supported requirement coverage and explicit gaps   | Must     | FR-10-12; NFR-01                          | Mapping remains within stated IDs and frozen truth yields per-requirement precision, recall, and macro-F1.               |
| Correct risky generated statements                          | Must     | FR-03, FR-06, FR-08-09                    | Stable claims expose unsupported, contradicted, unknown-citation, and altered-quote cases with reasons.                  |
| Reproduce a review packet                                   | Must     | FR-13, FR-15; NFR-02, NFR-08              | Clean fixture run reproduces normalized artifacts through the documented CLI/API path.                                   |
| Compare named models without a hidden score                 | Must     | FR-16-18; NFR-10                          | Two policies and optional label match; fail-closed incomplete compares write no <code>compare.json</code> .              |
| Operate within budget, privacy, and failure boundaries      | Must     | FR-01-02, FR-15; NFR-04-06                | Invalid, unsafe, over-budget, or provider-failed requests create no completed report and disclose no unapproved payload. |

## 6. Proposed product and innovation

The Workbench is a specialised evaluation pipeline, not a chatbot. A user validates one input bundle, evaluates named models, inspects retained reports, compares min-cost versus quality/task-fit, and appends a human review. The Inspector engine supplies claim-evidence audits; the compare shell does not replace it.

The AI component is deliberately replaceable. A **ModelGateway** interprets claim-evidence relationships; deterministic components control validation, budget and injection preflight, segmentation, retrieval provenance, evidence-integrity enforcement, metrics, artifacts, and review history. A provider response therefore cannot authorise an action or silently bypass evidence rules.

The engineering contribution is the observable linkage **claim -> evidence -> support decision -> requirement -> human review**. Unlike a generic critique prompt, the product has versioned schemas, stable claim IDs, retained citations and input hashes, bounded requirement IDs, typed model output, deterministic post-checks, no-clobber history, and explicit failure tests. It is independent from FActScore and RAGAs beyond cited context [4], [5]. This ELEC5623 product is independent new work; see **PRIOR\_WORK\_DISCLOSURE.md**. Tutor approval of that boundary is still pending.

## 7. Proposed methodology and system design

Requirement-led slices: acceptance criterion, shared service, pytest tests [7], evidence, traceability. The editable diagram names users, interfaces, services, data stores, external systems, the AI component, and the deployment boundary. The text pipeline is a supplement.

```
flowchart TB
    U[Users: student, evidence reviewer]
    I[Interfaces: CLI and FastAPI]
    S[Services: EvaluationService, ComparisonService]
    AI[AI component: ModelGateway]
    DS[Data store: local run directory]
    X[External: optional HTTPS model endpoint]
    DB[Deployment: Python 3.11 process + artifact root]
    U --> I
    I --> S
    S --> AI
    AI --> DS
    AI --> X
    S --> DB
```

CLI/HTTP are the interfaces. **ModelGateway** is the marked AI component and cannot write files or approve actions. Validation, citation/quote enforcement, safety preflight, logging, and human oversight remain outside it. Retrieval uses TF-IDF or a named lexical fallback [6].

```
JSON bundle -> schema/preflight -> segment -> retrieve -> ModelGateway
-> citation/quote checks -> metrics -> artifacts -> compare -> human review
```

Daily scene: one notebook, two AI write-ups. Validate, evaluate, compare, then human review. Success is an appended decision, not auto model-deploy.

| Field                     | Content                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor and goal            | Student/reviewer: which of two models hallucinates less / fits the notebook task better.                                                                                                                                        |
| Trigger                   | Two English write-ups for the same notebook; reviewer starts the Workbench.                                                                                                                                                     |
| Preconditions             | Valid English requirements, notebook evidence, both Markdown outputs; writable run root; fixture or approved gateway.                                                                                                           |
| Main success flow         | <b>validate</b> ; <b>evaluate</b> each model; <b>compare</b> quality/task-fit/time/cost and min-cost vs quality/task-fit; inspect reports; <b>review</b> appends a human decision. Status stays <b>READY_FOR_HUMAN_REVIEW</b> . |
| Alternate/exception flows | Invalid input, injection, budget, provider, mismatched truth, incomplete compare, or aliased artifacts fail before a completed report/ <b>compare.json</b> . Disagreement appends only.                                         |
| Postconditions            | Input/report immutable; reviews append-only; no model deployed or approved.                                                                                                                                                     |
| Linked FRs                | <b>validate</b> FR-01-03, FR-12; <b>evaluate</b> FR-04-15; <b>compare</b> FR-16-18; <b>review</b> FR-14; NFR-01, NFR-05, NFR-07, NFR-10.                                                                                        |

Design draft, not stakeholder evidence. Synthetic local data; CI uses **FixtureModelGateway**. Unknown citations and false quotes become **UNSUPPORTED**. **report.json** last. Limitations: lexical retrieval, English-only text, exact-string quotes, and no live-model generalisation.

## 8. Business and product analysis

The value proposition is a **reproducible human-review packet and a named-model comparison, not auto model-deploy**. Quality, task fit, latency, and estimated list-price cost stay separate; min-cost and quality/task-fit are contrasted, never collapsed. These benefits remain stakeholder hypotheses.

| Alternative                     | Strength                    | Gap for this proposed need                            | Workbench response                                               |
|---------------------------------|-----------------------------|-------------------------------------------------------|------------------------------------------------------------------|
| Manual review                   | Human context and authority | Structure and retained evidence may vary              | Preserve authority while standardising the packet                |
| Document search                 | Fast passage lookup         | No support classification or requirement map          | Add claim-level audit and bounded mapping                        |
| Generic LLM critique            | Low setup effort            | Free-form, unstable, and may invent evidence          | Strict schemas and deterministic enforcement                     |
| Research framework              | Established concepts        | Different task, data, metric, or workflow             | Cite context; implement independent requirements                 |
| Broad governance platform       | Operational breadth         | Potentially opaque, inaccessible, or too broad        | Deliver a narrow transparent semester slice                      |
| Token or model-price comparison | Easy to collect             | Silent on quality and task fit; can mislead selection | Keep dimensions separate; two named policies, not a hidden index |

Technical feasibility is supported by the Python vertical slice and fixture path; team feasibility is unknown until members and FR allocation are confirmed. Costs are engineering, annotation, approved model use if enabled, and runtime. No currency, customer, or savings claim is made. This is independent ELEC5623 Track B coursework.

Operational needs include Python onboarding, secrets outside source and artifacts, HTTPS for external credentials, storage retention, model-version records, provider outage handling, and ownership of the human queue. Authentication, tenancy, production support, and a commercial launch are out of scope. Stakeholder needs stay hypotheses until the proposed 20-minute tutor proxy walkthrough is agreed in writing and actually run.

## 9. Evaluation plan

This section is a **plan**, not a claimed final result. **fixture** and **fixture-b** are **development evidence that the software structure runs** (deterministic, no live network). They do **not** prove real-model effectiveness. Final product still needs a tutor-approved live model if allowed, a frozen corpus with annotation rules, and a real evaluation.

The evaluation asks whether traces stay complete (RQ1), five labels are distinguished (RQ2, quality), risky claims are detected (RQ3, quality), requirements are mapped (RQ4, task fit), failures close (RQ5, efficiency), results reproduce (RQ6), and min-cost vs quality/task-fit match annotation-preferred when labels exist (RQ7). Dimensions are not combined. Unlabelled compare is comparison only. Estimated cost is list-price × token bound, not a bill.

The planned dataset is 20 new ELEC5623 bundles / 200 claims. Before final evaluation: tutor approval, freeze label rules without seeing predictions, two permitted annotators or an approved alternative, independent disagreement resolution, hashed corpus/labels, no tuning on final labels. Current **fixture/fixture-b** numbers are development-structure evidence, not independent annotation.

Precision  $TP/(TP+FP)$ , recall  $TP/(TP+FN)$ , F1 their harmonic mean. Five-label macro-F1 averages class F1. Risky precision/recall treat **UNSUPPORTED** and **CONTRADICTED** as the positive set. Mapping macro-F1 is per-requirement then averaged. Citation completeness requires every cited ID to exist.

| Measure                        | Proposed target                                          |
|--------------------------------|----------------------------------------------------------|
| Citation completeness          | 1.00                                                     |
| Five-label macro-F1            | $\geq 0.65$                                              |
| Risky precision                | $\geq 0.80$                                              |
| Risky recall                   | $\geq 0.75$                                              |
| Requirement-mapping macro-F1   | $\geq 0.70$                                              |
| Elapsed time for 200 claims    | $\leq 60.0$ seconds on the nominated machine             |
| Two-model sample/daily compare | $\leq 60.0$ seconds; incomplete compares publish nothing |

These values are engineering proposals, not course thresholds, pending tutor review.

| Comparison                                            | Planned control                                             |
|-------------------------------------------------------|-------------------------------------------------------------|
| B0 trivial constant or frequency baseline             | Fit only on development data; report even if weak           |
| B1 TF-IDF, fixture gateway, deterministic enforcement | Reproducible Inspector baseline; list-price cost 0          |
| B1b gold-label fixture-b                              | Workbench contrast with published mock list price > 0       |
| B2 one compatible model                               | Out of compare CLI this week; privacy/course approval later |



Same-input ablations will remove TF-IDF, deterministic evidence enforcement, or the normal requirement-mapping signal. Failure tests cover malformed schema, duplicate or unknown IDs, missing or ghost ground-truth claim IDs, heading/code-only output, prompt injection, budget excess, provider timeout or invalid JSON, unknown citations, altered quotes, partial and reused artifacts, aliased report/review targets, and append-only review. Every unsafe failure must produce no completed report or outside-target change. Clean-checkout tests will reproduce the fixture run and verify normalized byte equivalence; timed performance will be measured on the nominated lab machine with runtime, hardware load, gateway, and command recorded.

The present local vertical slice has 152 passing tests and 94.02% branch coverage. The current no-clobber acceptance at [acceptance/local-20260804-review-integrity-v2/corpus-acceptance.json](#) reports 20 bundles and 200 claims: macro-F1 0.857701, risky precision/recall 0.930233/1.0, mapping macro-F1 0.96, and 37 label plus 12 mapping errors. Those 20/200 numbers remain labelled **DRAFT\_UNFROZEN development-fixture** evidence (**DRAFT\_UNFROZEN\_NOT\_TUTOR\_APPROVED**). They are not acceptance, not live-model effectiveness, and not a frozen result. The submission must report a later frozen evaluation separately with per-class metrics, errors, hashes, and all failed thresholds.

If permitted, at least one reviewer who did not author the evaluated bundle will locate a risky claim, inspect evidence and requirement traces, append a decision, and judge packet sufficiency. Record task completion, correctness against frozen annotation, time, missing fields, and a short usefulness/trust rating while keeping identity and comments private. If unavailable, use the tutor-approved qualitative alternative. Validity threats include lexical and exact-quote proxies (construct), generator-fixture alignment (internal), 20 English synthetic bundles (external), aggregate-only interpretation (conclusion), and environment drift (reproducibility).

## 9.1 Evaluation trace and interpretation rules

| Research question                               | Primary evidence                                                              | Candidate decision rule                                                                                               | Required report content                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| RQ1: Are claim-evidence traces complete?        | Schema checks, citation completeness, manual packet sample                    | Target 1.00; any missing required trace field is a failure, not a partial pass.                                       | Missing fields by claim and producing stage                                 |
| RQ2: Are five support labels distinguished?     | Frozen confusion matrix and per-class metrics                                 | Proposed macro-F1 $\geq 0.65$ ; also report every class even if the aggregate passes.                                 | Per-class precision/recall/F1 and confusion matrix                          |
| RQ3: Are risky claims surfaced?                 | Combined unsupported/contradicted precision and recall                        | Proposed precision $\geq 0.80$ and recall $\geq 0.75$ ; neither metric may hide the other.                            | False-support and false-escalation examples                                 |
| RQ4: Are requirements mapped accurately?        | Frozen per-requirement truth and macro-F1                                     | Proposed macro-F1 $\geq 0.70$ with unknown IDs rejected before scoring.                                               | Per-requirement errors and uncovered requirements                           |
| RQ5: Does the product fail closed?              | Negative suite for injection, budget, provider, evidence, and artifact faults | Every unsafe case produces no completed report; any readable partial success fails.                                   | Input, expected failure, observed artifact state, and diagnostic            |
| RQ6: Can the result be reproduced and reviewed? | Byte-equivalent fixture runs, clean checkout, and permitted reviewer task     | Exact fixture equivalence plus documented environment; qualitative findings remain separate from quantitative claims. | Runtime/configuration, hashes, task outcome, time, and missing trace fields |
| RQ7: Do named policies agree with labels?       | Two-model compare on a labelled bundle                                        | Min-cost vs quality/task-fit vs annotation-preferred; unlabelled runs set verification unavailable.                   | Selected models, costs, quality metrics, and match flags                    |

Threshold outcomes will be reported as pass or fail without suppressing failed classes, scenarios, or requirements. Error analysis will select representative false support, false escalation, mapping error, citation/quote enforcement, and provider-failure cases using a rule fixed before the final run. The group will link every reported table or figure to a retained run ID and source hash so the proposal, later report, demo, journal, and commit can be reconciled.

## 10. Risks, ethics and responsible AI

NIST AI RMF organises AI risk work through GOVERN, MAP, MEASURE, and MANAGE and identifies accountability and transparency as trustworthiness concerns [1]. Its GenAI Profile discusses confabulation, privacy, human-AI configuration, automation bias, information integrity, and security [2]. These references guide questions; they do not prove that this prototype is safe or compliant.

The main ethical risk is presenting a support label as truth. The interface and report therefore expose evidence and uncertainty, keep **READY\_FOR\_HUMAN\_REVIEW** as the only successful machine state, and record human decisions separately. The demo must include a human challenge to an automated assessment. Both risky precision and recall are reported because false support may hide a weak claim while false escalation may waste effort or unfairly undermine a valid one.

Synthetic data is the default until provenance, consent, licence, retention, and approval are recorded. Secrets stay outside source and evidence. OWASP describes direct and indirect prompt injection and layered controls including output validation, least privilege, untrusted-content separation, adversarial testing, and human approval [3]. The current pattern guard is not foolproof; the stronger control is that the gateway cannot write artifacts or act externally. Bias can enter domains, labels, annotation, and per-class performance, so errors will be reported by class. Data rights, provider terms, assessment rules, queue ownership, over-reliance, and onboarding remain legal or adoption gates. No production security or regulatory compliance is claimed.

| Risk                                              | Current mitigation                           | Trigger and response                      | Owner gate                   |
|---------------------------------------------------|----------------------------------------------|-------------------------------------------|------------------------------|
| Tutor or prior-work decision conflicts with scope | Early approval pack and disclosure           | Stop freeze; revise and reconfirm         | [REQUIRED: authorised tutor] |
| Team availability differs                         | Conditional allocation and shared review     | Replan without inventing members          | [REQUIRED: actual group]     |
| Synthetic corpus overfits fixture rules           | Independent protocol and frozen labels       | Report limitation; redesign evaluation    | [REQUIRED: evaluation owner] |
| Provider leaks data or fails                      | Synthetic minimum payload, HTTPS, fixture CI | Fail closed; use approved fixture path    | [REQUIRED: technical owner]  |
| Automation bias                                   | Human-only decision and challenge case       | Record disagreement; revise presentation  | [REQUIRED: whole group]      |
| Requirement or schedule drift                     | Traceability, hashes, weekly gates           | Block report/demo; cut Should/Could first | [REQUIRED: project owner]    |

### Direct GenAI and prior-work disclosure

OpenAI Codex was materially used on 2 August 2026 to scaffold and review the independent prototype, tests, evaluation tooling, and this proposal candidate. Its outputs were revised and checked using local tests, source provenance, and separate Codex-assisted adversarial review passes; no independent human review is implied. Codex does not establish stakeholder facts, team authorship, tutor approval, frozen-corpus validity, or course compliance. The submitting group remains responsible for every claim, citation, requirement, engineering decision, and permitted-use declaration.

**AI\_JOURNAL.md** records detailed use, but does not replace this direct acknowledgement. Before submission the group must reconcile the wording with current course and University guidance, record all later material AI use and human review, and retain the tutor's decision on **PRIOR\_WORK\_DISCLOSURE.md**.



## 11. Semester plan and team responsibilities

| Period                                             | Intended outcome and exit evidence                                                                                    | External dependency                              |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 3-9 Aug                                            | Form/register group; confirm pitch, schema, 18 FRs, and prior-work disclosure                                         | Members and Canvas group rule                    |
| 10-16 Aug                                          | Complete Lab 1 environment evidence and single-bundle vertical slice                                                  | Private approved accounts and course environment |
| 17-23 Aug                                          | Complete Week 3 oral check; record feedback and scope decision                                                        | All-member participation and tutor feedback      |
| 24-30 Aug                                          | Approve corpus protocol; implement baselines, failure tests, and draft evaluation                                     | Tutor/annotation decision                        |
| 31 Aug-7 Sep                                       | Obtain formal approval; reconcile 8-10 page PDF, citations, AI use, and all versions                                  | Tutor, group review, live Canvas check           |
| 8 Sep 23:59                                        | Student submits one group PDF and retains confirmation privately                                                      | Student action                                   |
| 9 Sep-late Sep                                     | Implement the approved vertical slice, integrate Must interfaces, and convert new Canvas criteria into traceability   | Published assessment/rubric updates              |
| Late Sep-mid Oct                                   | Reach the internal MVP target; close Must tests and run failure scenarios without expanding approved scope            | Team integration and approved environment        |
| Mid-late Oct                                       | Reach the internal feature-complete target; freeze the permitted test set and produce the report/error-analysis draft | Frozen protocol and later brief                  |
| Final two teaching weeks before the published demo | Feature freeze, clean-checkout reproduction, report/AI-journal audit, and at least two demo/Q&A rehearsals            | Official final-assessment instructions           |
| Official final demonstration date once published   | Demonstrate the version matched by report, journal, evidence, and final commit                                        | Canvas date and in-person assessment             |

For a confirmed five-person group, the neutral allocation is Slot A FR-01-03, Slot B FR-04-06, Slot C FR-07-09, Slot D FR-10-12, and Slot E FR-13-18. For fewer members, all 18 FRs must be redistributed while preserving at least three per person. Replace slots with **[REQUIRED: actual names, SIDs, availability, primary deliverables, and reviewers]** only after agreement. Every member remains responsible for the complete proposal.

The group will log each requirement change with source, owner, and impacts on implementation, tests, evidence, and report. A feature needs matching acceptance evidence and review. Proposal, code, tests, demo, AI journal, and final commit must describe one version. Must defects outrank Should/Could features, and private course or participant data stays private.

Before rebuilding this candidate as the submission PDF, the group must close these gates: live Canvas rule recheck; group registration and allocation; Week 3 participation; written tutor approval and cover details; stakeholder or approved alternative evidence; prior-work decision; annotation and threshold decision; references and AI-use audit; final internal consistency and member sign-off. The student/group alone performs the final Canvas submission. No appendix allowance is assumed.

## 12. References

- [1] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, Jan. 2023, doi: [10.6028/NIST.AI.100-1](https://doi.org/10.6028/NIST.AI.100-1).
- [2] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*, NIST AI 600-1, July 2024, doi: [10.6028/NIST.AI.600-1](https://doi.org/10.6028/NIST.AI.600-1).
- [3] OWASP Gen AI Security Project, "LLM01:2025 Prompt Injection," accessed 2 Aug. 2026. <https://genai.owasp.org/llmrisk/llm01-prompt-injection/>.
- [4] S. Min, K. Krishna, X. Lyu, M. Lewis, W.-t. Yih, P. Koh, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi, "FactScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation," in *Proceedings of EMNLP 2023*, pp. 12076-12100, 2023, doi: [10.18653/v1/2023.emnlp-main.741](https://doi.org/10.18653/v1/2023.emnlp-main.741).
- [5] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated Evaluation of Retrieval Augmented Generation," in *Proceedings of the 18th EACL: System Demonstrations*, pp. 150-158, 2024, doi: [10.18653/v1/2024.eacl-demo.16](https://doi.org/10.18653/v1/2024.eacl-demo.16).
- [6] scikit-learn developers, "TfidfVectorizer," *scikit-learn API Reference*, accessed 2 Aug. 2026. [https://scikit-learn.org/stable/modules/generated/sklearn.feature\\_extraction.text.TfidfVectorizer.html](https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html).
- [7] pytest developers, "pytest documentation," accessed 2 Aug. 2026. <https://docs.pytest.org/en/stable/>.

All seven references were opened and metadata-checked on 2 August 2026. They support design context only and do not prove implementation performance, safety, compliance, stakeholder demand, tutor approval, or a frozen result.

NOT FOR SUBMISSION